Chapter 14

Provisional restorations

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INTRODUCTION

During the time period between tooth preparation and fit of an indirect restoration it is important in most situations to provide a patient with a high-quality provisional restoration. Failure to do so could lead to the early demise of the definitive restoration for reasons which will be outlined in this chapter. This comes at great expense to both patient and dentist, in terms of both monetary and biological cost, and professional relationship and trust.

Some may argue, why waste precious time crafting a bespoke provisional restoration only for it to be replaced within weeks? Moreover, the cynic would argue that acceptance of the definitive restoration can be problematic if the dental aesthetic and function of the provisional restoration is comparable to that of the final restoration. To the contrary, when the dentist appreciates the relationship between form and function of a provisional restoration and its relationship to the immediate and long-term health of the teeth, supporting structures and the definitive restoration, its importance is unquestioning; the dentist/health care professional now only has to acquire the knowledge and skills to construct one.

Oral health care and treatment planning for advanced operative procedures follows a logical sequence as outlined in the earlier chapters of this textbook: initially stabilizing (including preventing further) dental diseases, evidence of oral health and then the possible reconstruction of teeth. The success of each stage of management depends upon the success of the preceding stage. The same is true when providing laboratory-fabricated indirect restorations. It is unacceptable to provide a laboratory-fabricated restoration without the prior placement of a provisional restoration. Whilst it is true that a less than ideal provisional restoration might not always influence long-term outcome of the definitive restoration, it may result in a lengthy 'fit appointment'. A health care professional would not be discharging their responsibility if any aspect of their treatment is substandard, including the placement of provisional restorations. The synergy between an empowered patient, dentist and dental technician can and should provide a seamless continuum of dental care and provisional restorations are a part of this.

With respect to terminology, there is a facile debate as to whether or not such restorations should be called

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provisional or temporary. The debate is perhaps academic and in this section they will be referred to as provisional restorations.

CHARACTERISTICS OF A PROVISIONAL RESTORATION

When considering provisional restorations it is most logical to consider features that must be achieved and then other 'value-added' functions of the restoration. Before exploring these, at the centre of every carer's ethic is '*Primum nil nocere*' (First, do no harm). Preparing a tooth for a laboratory-fabricated restoration will, however, by necessity:

- Compromise its dental aesthetic if in the smile line
- Degrade the tooth's function as a result of occlusal reduction
- Result in an unstable occlusion due to occlusal and approximal reduction
- Render a vital tooth sensitive due to unprotected freshly cut dentine
- Compromise the coronal seal to root-filled teeth.

A provisional restoration should therefore restore the characteristics that have been lost and additionally allow optimum home care. 'First, do no harm' also applies to something seemingly as trivial as providing a provisional restoration. Consider the provisional crowns provided in Figures 14.1 and 14.2. The provisional restoration in Figure 14.1 demonstrates an adequate fit cervically and home care should therefore not be compromised. In contrast, the provisional restoration that has been placed on the molar tooth in Figure 14.2 had overhanging margins and during the short time in situ adequate oral hygiene has not been possible and gingival inflammation has ensued.

Specifically, therefore, a provisional restoration must:

- *Establish and/or maintain the dental aesthetic*. A 'value-added' function of a provisional restoration would be to evaluate a patient's satisfaction with the dental aesthetics if this is to be changed in the definitive restorations. A diagnostic wax-up can be made of the ideal aesthetics for the tooth/teeth. This can then be duplicated in stone to allow a vacuum-formed splint to be made which can then be used to make a custom-formed temporary crown(s) (see later).
- *Confirm that the tooth preparation is adequate*. This includes sufficient occlusal reduction and intra- and extracoronal design characteristics, including maintaining occlusal stability (the prepared tooth must not be allowed to drift or overerupt in an uncontrolled fashion). A 'value-added' function of a provisional restoration is as a tool to help the dentist maintain a conformative approach to the occlusion.



Figure 14.1 This provisional restoration demonstrates an adequate fit, cervically. Home care should therefore not be compromised and periodontal health maintained.



Figure 14.2 A poorly fitting provisional crown has been removed from the upper left first molar tooth. The overhanging margins mesially and palatally have prevented adequate home care with resultant plaque accumulation and gingival inflammation.

- *Prevent tooth sensitivity* by covering all exposed, freshly cut dentinal tubules in vital teeth and preventing fluid movement in the dentinal tubules.
- *Prevent bacterial leakage* at the provisional/tooth tissue interface in vital teeth and non-vital, root-filled teeth. The former reduces pulpal inflammation and

the latter maintains a good coronal seal to the root canal filling.

- *Allow optimum home care*. Ledges are unacceptable, but anecdote would suggest it is preferable to have a negative (tooth) ledge when compared with a positive (restoration) ledge (Figure 14.2).

Provisional restorations can be either custom-formed to each individual situation or preformed by manufacturers in standard shapes and sizes and adjusted to fit at the chairside. The custom-formed temporaries are preferred, but are perhaps a more demanding technique to master.

CUSTOM-FORMED RESIN REPLICA PROVISIONAL CROWNS

The most appropriate material to be used for a custom-formed resin replica provisional restoration is a chemically cured bis-acrylic composite resin, for example Protemp Plus Temporisation Material (3M ESPE) or Integrity Temp-Grip Temporary Crown and Bridge Material (Dentsply). The merit of using this as a provisional restorative material is that it can be customized so that its internal aspect custom fits the preparation and its external surface reproduces accurate contact points and occlusion with the opposing arch.

It is important that the provisional restoration is made before the impression is taken. This is because the provisional restoration can be used to affirm that the tooth preparation characteristics are satisfactory. These include whether sufficient tooth/core has been prepared to accommodate the definitive restoration (Figure 14.3) and other preparation characteristics such as whether the preparation is undercut or not (Figures 14.4 and 14.5). If deficiencies in the preparation are found these can be remedied and the temporary can be relined or remade. An impression of the preparation can now be made with reassurance that the preparation is satisfactory.



Figure 14.3 An Iwanson gauge can be used to measure the thickness of the provisional restoration in relation to the buccal cusp occlusal reduction, for example. If it is shown that there is insufficient tooth reduction, further preparation is carried out and the provisional restoration is relined.



Figure 14.4 Bis-acrylic composite provisional restoration, affirming a satisfactory preparation for a gold overlay. This provisional restoration shows mesial and distal gingival bevels to the proximal box preparation.



Figure 14.5 The provisional restoration shown in Figure 14.4 in situ.

How to make a custom-formed provisional restoration (Figure 14.6)

Firstly a sectional impression of the tooth to be prepared is made. A full arch impression is unnecessary for this and would make it more difficult to relocate on the teeth. This can be done in a number of materials:

- An addition cured silicone putty is preferred as this can be disinfected and given to the patient (Figure 14.6). If the temporary is lost or breaks in function in the ensuing weeks whilst the crown is being made, the impression can be reused (as it is dimensionally stable) to make a new custom-formed



Figure 14.6 The stages involved in making a custom-formed resin replica provisional crown. From left to right and top to bottom: an impression is taken of the tooth prior to preparing it for a crown. The tooth is then prepared, the impression of the tooth is filled with a bis-acryl resin and reseated fully in the mouth. Once the bis-acryl resin has almost set (rubbery stage) the impression is removed. The provisional restoration is removed from the tooth and trimmed.

provisional restoration. Such a material has the advantage over alginate, in that it can be used repeatedly without tearing.

- Alginate is cheaper, but cannot easily be reused or kept by the patient should the provisional restoration fail in function.
- Some clinicians use softened modelling wax. This has the advantage that it can easily be adjusted and smoothed with a hot instrument (see later) but is not suitable where there are deep undercuts as it would easily distort. Whilst cost effective and convenient to make, it cannot be stored and used again, unlike the silicone matrices. For this reason the technique is suitable for simple and single restorations that copy the original shape of the tooth.
- A custom vacuum-formed plastic mould can be made from a study model or a model made from a diagnostic wax-up (see later).

The tooth is prepared for the chosen design of restoration, which in Figure 14.6 is a full coverage gold crown. A thin layer of petroleum jelly can be smeared onto the preparation to facilitate the removal of the provisional restoration from the tooth once set; this is not always necessary. The bis-acrylic composite resins used are supplied in cartridges which fit into dispensing guns onto which double helix mixer tips are attached. Before syringing the material into the impression, some of the material should be dispensed onto the bracket table or a mixing pad to ensure it is completely mixed (occasionally, base or catalyst moves into the double helix mixer tip ahead of the other and hence emerges in an unmixed state) and to monitor its set. The bis-acrylic composite resin is then syringed into the sectional impression of the tooth that has been prepared and the impression is relocated in the mouth, ensuring it is fully seated over the teeth; an obvious click is often felt as the impression passes over the bulbosity of the remaining teeth.

Before complete polymerization of the bis-acryl resin, remove the impression. At this stage the resin on the pad or bracket table will feel rubbery. This is an important step as waiting for the final set may result in difficulty in removing the temporary restoration as it can set into existing undercuts. For this reason at the rubbery stage carefully move the matrix up and down with a small amplitude so as to ensure no setting occurs into such undercuts. The provisional restoration that has formed may either stay on the tooth or will be removed in the impression. In the former situation, gently ease it off the preparation as soon as possible using an instrument beneath the contact points. If this is not done, the bis-acrylic composite will completely polymerize into the interdental undercuts, making it impossible to remove without destroying the provisional restoration. If the provisional is removed in the impression, let it set completely as trying to remove it in a partially polymerized stage could lead to distortion and damage.

Once removed from the mouth or impression, any material flash and ledges can be removed with a high

speed diamond bur or abrasive polishing discs, paying particular attention to the interdental areas where the material goes into the undercut areas beneath the bulbosity of the adjacent teeth. Following this, confirmation of an adequate tooth preparation can be carried out as described earlier and in Figures 14.3 and 14.4, comparing the fit surface of the provisional restoration with that of the prepared tooth. The marginal fit and occlusion are checked with the provisional restoration in situ and adjusted if necessary; ideally, the provisional restoration should be removed and adjusted outside of the mouth. Finally, the patient should be shown the restoration to confirm that the dental aesthetic is satisfactory. The provisional restoration can then be cemented with a temporary luting cement such as a non-eugenol temporary cement (TempBond NE, Kerr Dental). It is argued that eugenol-containing cement would inhibit the polymerization of a permanent resin luting cement. Excess temporary cement is then meticulously removed from the margins, carefully using dental floss interdentally. To avoid dislodging the provisional restoration by pulling the floss back through the contact point, the floss can be pulled out buccally/labially.

Modifications to the technique

Occasionally it is necessary, before preparing the tooth, to modify its shape with soft wax or resin composite (placed and cured but not etched and bonded to the tooth or core) to achieve a better contour for the provisional restoration. This is necessary in situations where the core or shape of the tooth is inadequate. For example, the amalgam core seen in Figure 14.7 has no disto-buccal cusp created in the core and it is not in occlusal contact in this area with the opposing teeth. Minimal or no occlusal reduction is needed in this area but using the custom-formed technique for making provisionals as described above would lead to a very thin layer or no provisional material in the unprepared area and a corresponding perforation in the provisional restoration would result. To address this, wax or composite can be placed to build up the cusp to form and function before the preoperative impression is taken, ensuring the provisional restoration is intact and correctly contoured.



Figure 14.7 Poorly contoured amalgam restoration which is to act as a core for a full coverage crown. The occlusal surface can be modified using composite or wax to create the ideal occlusal form, prior to taking an impression for a custom-formed resin replica provisional restoration.

An alternative method to overcome such problems is that the impression itself can be adjusted in the relevant areas using a scalpel blade or instrument such as an excavator. The amount of impression material removed is arbitrary and the provisional restoration often needs more adjustment. Advocates of using softened wax as an 'impression material' claim that the wax is easier to remove and can be smoothed with a hot instrument.

Establishing the occlusion and aesthetics with custom-formed provisional restorations

When placing crowns and/or bridges on anterior teeth which have lost their original form, for example through tooth wear or repeated restoration over many years, the shape of the teeth needs to be re-established; the occlusion and aesthetics can be piloted on provisional restorations. Ideally, guidance (anterior/incisal) should be produced on the crowns and this can be created in a diagnostic wax-up on articulated study models. Once this and the appearance are satisfactory, the wax-up can be duplicated in stone and a vacuum-formed mould (template/splint) made. At the patient's subsequent visit, the anterior teeth can be prepared and the mould used to make custom-formed provisional restorations to the new occlusal scheme and appearance created in the diagnostic wax-up. The patient can then wear the provisional restorations for some time until happy with the form and function (Figure 14.8). During this time the occlusion and appearance (shape of the teeth) can be altered by removing provisional material with a bur or the addition of a colour-matched composite, if necessary.

The guidance created in such provisional restorations can be transferred to the definitive restorations using a technique such as a customized formed incisal guidance table (Figure 14.9). In this technique, impressions of the upper (including the provisional restorations) and lower arches are made and the resultant models mounted on a semi-adjustable articulator as described in Chapter 6. Lateral and protrusive movements are reproduced on the articulated models. As this is done the incisal pin raises up off the incisal guidance table according to the steepness of the guidance created on the provisional restorations. The technician then places cold cure acrylic on the incisal guidance table and repeats the excursive movements, the



Figure 14.8 Establishing incisal/anterior guidance on the provisional restorations. Note 'light' if any contact on pontic upper right lateral incisor to minimize unfavourable forces during protrusion and excursion.

incisal guidance pin moulding (scoring) the setting acrylic. The so-called customized incisal guidance table is therefore formed.

Impressions of the tooth preparations are then made and the resultant master model mounted on the articulator to the lower model. As the definitive restorations are being made, the technician repeatedly makes the excursive movements, now guided by the custom-formed incisal guidance table. The palatal contours of the crowns are adjusted to have simultaneous contact with the incisal pin on the guidance table, so reproducing the guidance of the provisional restorations in the definitive crowns.

Fortunately for most circumstances a diagnostic wax-up produced by a technician can satisfy the aesthetic demands of a patient. Using the guides described in Chapter 15, the form and shape of restorations can be made to complement the mouth. However, on occasion, some patients can have a high aesthetic demand. In these situations the initial provisional crown may be altered or replaced to achieve the demands of the patient. Chairside alterations, provided they are minor, can be undertaken by removing and reshaping with burs or adding small amounts of provisional material. More extensive changes may need a replacement provisional restoration; this is more likely on upper anterior teeth where the aesthetic demands of most people are higher. After agreement on the restoration shape, an alginate impression can be recorded to help the technician reproduce the result in the final restoration.

Establishing gingival contours with custom-formed provisional restorations

In certain situations custom-formed provisional crowns can allow shaping of the gingival tissues in order to achieve a satisfactory emergence profile of the definitive restoration. This is indicated particularly when restoring bone-level dental implants. For details on this the reader should refer to a text on implantology.

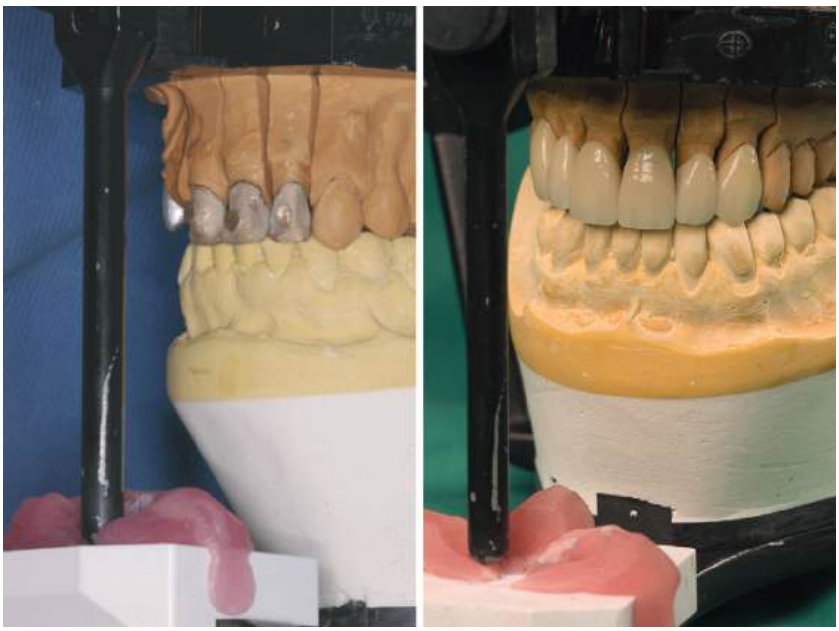


Figure 14.9 Customized incisal table replicating the palatal contours of those established in the provisional restorations made in Figure 14.8 (left), and for another patient whose guidance is being replicated in replacement crowns (right).

PREFORMED PROVISIONAL CROWNS

A number of preformed provisional crowns are available. These can be tooth coloured (polycarbonate crowns) or clear plastic crown forms which can be filled with bis-acryl composite resin (for use within the smile line), or they can be metal based. Each type of crown is provided as a series with different tooth morphologies and sizes. This poses two problems: (1) it is unlikely that a preformed crown will fit cervically, occlusally and interdentally with any degree of accuracy and, as such, will require chairside adjustment; and (2) a large bank of crowns (at a monetary cost) is required to fit any eventuality. This having been said, they are particularly useful in situations where an impression of the tooth prior to crown preparation does not exist, for example when a patient presents following trauma and significant loss of coronal tooth tissue.

Provisional polycarbonate crowns

Polycarbonate crowns (e.g. Directa, JS Dental Manufacturing Inc.) are tooth-coloured 'shells' which have a morphology to meet all anterior and bicuspid teeth (Figure 14.10). These products have been available for more than 35 years and in use are sublined with an acrylic such as Trim (PEMA temporary resin acrylic). First, a polycarbonate shell is selected that is slightly larger than the preparation dimension (Figure 14.11). A bur such as a large pink stone in a straight handpiece is then used to pare it back until it is of the correct preparation dimension and seats fully over the tooth preparation without bedding into the gingiva. A thin smear of petroleum jelly is applied to the tooth preparation and the acrylic resin (Trim) is spread into the fit surface of the crown. This is then seated over the preparation and positioned such that the aesthetic is satisfactory. When the acrylic has polymerized to a granular/dough stage, the shell is carefully removed and replaced, before finally removing to allow the acrylic to polymerize fully outside the mouth. The provisional restoration is then trimmed until the fit is acceptable (Figure 14.12). If the crown is overbuilt, blanching of the gingiva will occur and further trimming is indicated. Following impression taking, the provisional crown can be cemented with a temporary luting cement.



Figure 14.10 Polycarbonate Directa crowns.



Figure 14.11 Adequate definitive restorations spawn from satisfactory tooth preparations and provisional restorations. These are the preparations for the provisional restorations as seen in Figure 14.12.



Figure 14.12 Adequate provisional restorations on the upper right central and lateral incisor made from polycarbonate Directa crowns. The buccal contour of the provisional restoration on the upper left central incisor is poor.

Other provisional restorations

In a similar way to the use of the polycarbonate crowns, clear plastic crown forms (Figure 14.13) can be selected and trimmed until they seat fully over the tooth preparation without traumatizing the gingiva. These are thin and can easily be trimmed with a scissors. Prior to filling with a bis-acryl composite resin, small holes can be made with a probe at the cusp/canine tips and incisal angles.



Figure 14.13 Clear plastic crown forms, suitable for use with bis-acryl composite resin.

This allows air to escape, reducing the risk of bubbles forming within the resin. Once the bis-acryl composite resin has set, the clear plastic crown form can be removed, the margins and occlusion can be checked and adjusted as necessary, and the crown cemented with a temporary cement.

Metal aluminium and stainless steel provisional crowns are also available for use on posterior teeth (Figure 14.14). Their margins are more difficult to adjust and ledges and sharp margins are more likely to cause soft tissue trauma (Figure 14.15). Some metal provisional crowns are provided with a crimping device that can be used to mould the margins of the crown to the tooth shape.

Occasionally, if replacing a crown, the original crown can be modified and used as a temporary. Consider the crown in Figure 14.16 which is being replaced because of distal caries. It has been partially sectioned, and following caries removal and refinement of the preparation, has been relined with a bis-acrylic composite resin and cemented temporarily.

If the original crown is planned to be used as a provisional restoration, consideration should be given to its removal with a WAMkey (WAM). The WAMkey system is



Figure 14.14 Metal provisional crowns.



Figure 14.15 A poorly trimmed and contoured provisional aluminium crown. The margins are sharp and there is a ledge mesially.



Figure 14.16 The partially sectioned crown on the molar tooth has been relined with a bis-acryl composite resin and cemented temporarily.

a set of three elevators, each with an oval end of increasing diameter. A hole is cut in the buccal aspect of the crown at the interface between the occlusal surface of the core and occlusal surface of the crown (Figure 14.17). The appropriate sized WAMkey is inserted at the interface and rotated, elevating the crown off the preparation with minimal risk of damaging the core and tooth preparation. Crown and bridge removers, such as pneumatic crown and bridge removers, that send a shock wave through the cement lute to shatter it, should be avoided as these can lead to decoronation of the tooth.

More recently, preformed malleable composite crowns have been introduced (Protemp Crown Temporisation Material; Figure 14.18). These are soft and easily moulded to the tooth preparation in situ. Once the desired shape is achieved they can be partially light cured in the mouth. It is important to only partially light cure the crown in the mouth for about 2–3 seconds as complete cure can lead to difficulties in their removal from the tooth. The crown



Figure 14.17 Use of a WAMkey to elevate off a metal–ceramic crown which has caries distally. The cast post and core beneath remain intact following caries removal and preparation refinement (Inset).



Figure 14.18 Protemp provisional crowns.

is removed and fully cured outside of the mouth. Final check of fit and any adjustment can be carried out prior to cementing with a temporary luting cement.

PROVISIONAL REPLACEMENT OF MISSING TEETH

Where full preparation, conventional bridges are concerned, provisional restorations can be made in a similar manner to custom-formed provisional crowns, using a diagnostic wax-up of the replacement tooth. Alternatively, a simple acrylic removable partial prosthesis can be used. This can be used if a minimum preparation bridge (resin retained) or implant retained restoration is planned; if adjacent teeth are to be prepared for a conventional bridge, the removable prosthesis can be used in conjunction with provisional crowns.



Figure 14.19 The upper right central incisor was restored with a post crown that catastrophically failed. This tooth was extracted and a bone-level implant placed 8 weeks later. Note staining due to the use of chlorhexidine following surgery.



Figure 14.20 Ethylene vinyl acetate (EVA) copolymer splint with acrylic tooth replacing the upper right central incisor, secured in place with an adhesive.

Another method of provisionally replacing a missing tooth where no preparation of the adjacent teeth is planned (Figure 14.19) is to take an impression and place an acrylic prosthetic tooth in the edentulous space on the model. A plastic (ethylene vinyl acetate (EVA) copolymer) splint can then be made over this and the adjacent teeth (Figure 14.20). The acrylic tooth can then be secured to the splint using an adhesive, giving an acceptable appearance (Figure 14.21).

ADVICE DURING THE PROVISIONAL STAGE

A patient should be urged to disrupt regularly the plaque biofilm associated with provisional restorations, enlisting the full armamentarium of cleaning aids. One notable



Figure 14.21 The acrylic tooth in clear splint has adequate aesthetics for a provisional tooth replacement (for patient seen in Figure 14.19). Oral hygiene has to be good.

caution is the use of dental floss which could pull out a provisional restoration if pulled back up through the contact point (it should be pulled out buccally). However, if the patient does not maintain adequate home care, gingival inflammation will result in an increased volume of gingival crevicular fluid or even gingival bleeding. At cementation this could lead to an inadequate cement lute. It is therefore good practice to advise patients to continue with routine dental home care.

SUMMARY

A carefully made provisional restoration should have all the characteristics of a definitive restoration. Unequivocally it is a surrogate marker for a dentist in determining the appropriateness of the tooth preparation and identifying errors. Custom-formed provisional restorations are also important when changes are planned to the appearance and occlusion. The demand for a quality provisional restoration is frequently overlooked in the focus to perfect tooth preparation and ensure an accurate impression; this is an error that should be avoided at all costs.

FURTHER READING

Wassell, R.W., St George, G., Ingledew, R.P., Steele, J.G., 2002. Crowns and other extra-coronal restorations: provisional restorations. Br. Dent. J. 192, 619–622, 625–630.